Big Data: End-to-End Linear Regression Pipeline

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Introduction

When data became cheaper and easier to store, it created a new problem. Analysts no longer suffered from data scarcity, but handling the massive datasets became more cumbersome. Luckily, many solutions have been developed to address this problem, and this project demonstrates how multiple solutions can be used within a single big data pipeline.

Dataset

Since the project focuses on the pipeline itself and was deployed on Google Cloud's free-trial virtual machine, a small synthetic dataset was used to reduce compute costs, while still demonstrating the basic concepts. The dataset was generated by AI using ChatGPT and contains the following variables: student_id, sleep_hours, gpa, and bedtime_category (early, moderate, or late). The goal of the pipeline is to simulate a big data approach to predicting students' GPAs using sleep-related features.

Pipeline Overview

The pipeline will use six software components, each designed to serve a distinct purpose within the big data pipeline: NiFi, HDFS, Apache Hive, Apache Spark, HBase, and YARN. NiFi will be used for data ingestion, HDFS for data storage, YARN for resource management, Hive for data warehousing, HBase for NoSQL storage, and Spark for data processing, querying, and machine learning. The data will be ingested via NiFi, loaded into HDFS, loaded into a Hive table, analyzed in Spark using SQL and MLlib, and the final evaluation metrics will be written to HBase for easy querying.

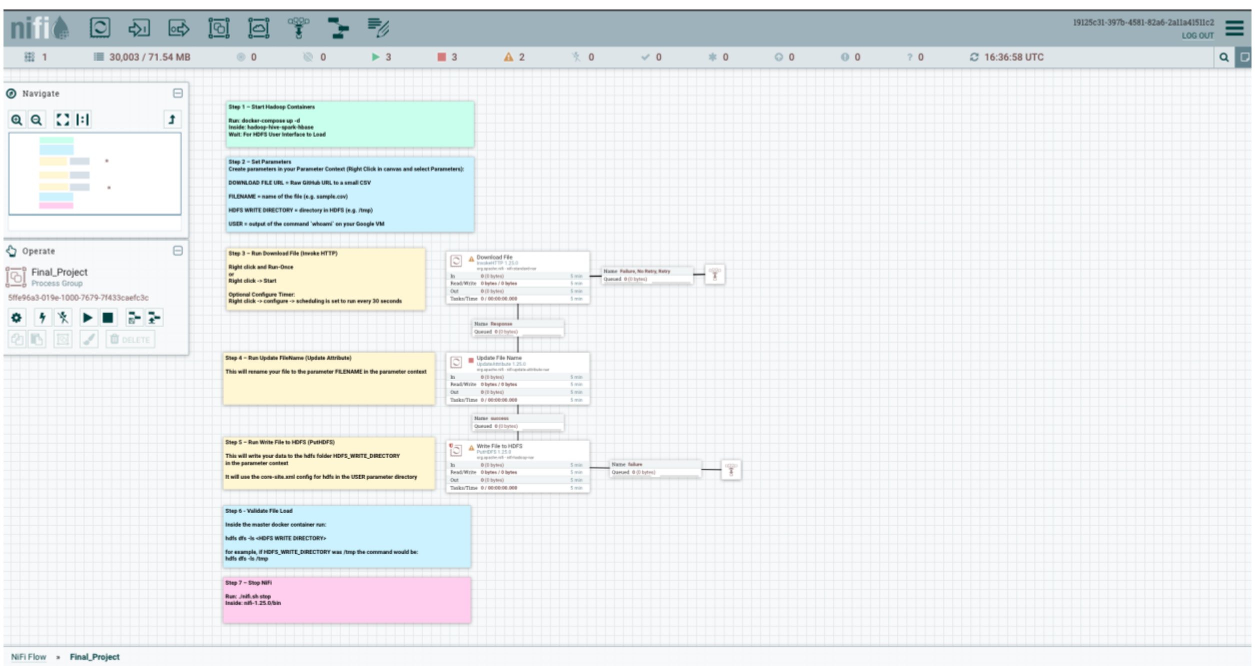
Issues Encountered

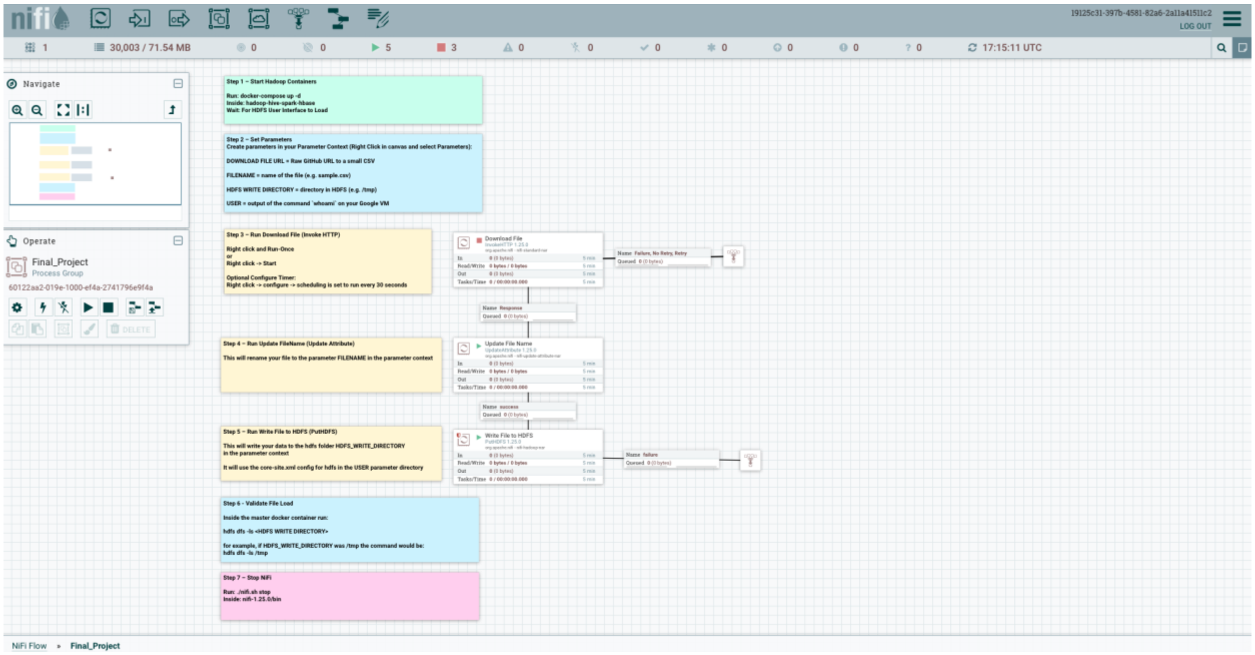
While the pipeline was successful end-to-end, an error occurred when I initially loaded the data into HDFS via NiFi. The process showed a single error within the queue. On further investigation, it was discovered that the error occurred because the file had already been loaded into HDFS. An extra click must have caused the data to be ingested more than once, causing the error. It was easy to remove the error from the queue, and the verification command output proved that the file had been loaded to HDFS without issue.

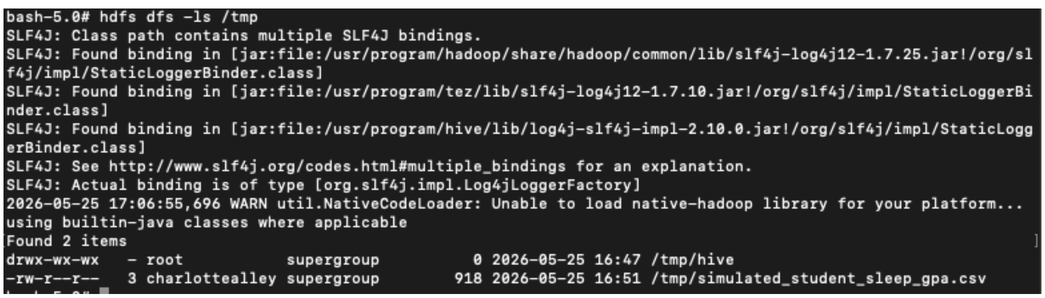
Screenshots & Code

NiFi Data Ingestion into HDFS

This NiFi flow uses InvokeHTTP, UpdateAttribute and PutHDFS to ingest the raw github file into HDFS. InvokeHTTP retrieves the file from the GitHub raw URL that was entered when defining the parameters, and brings it into NiFi as a FlowFile. UpdateAttribute modifies the filename to the specified name, also established in the parameters, and PutHDFS writes the FlowFile to the targeted HDFS directory. The file was successfully written into the HDFS directory, as confirmed by the hdfs dfs -ls command output.







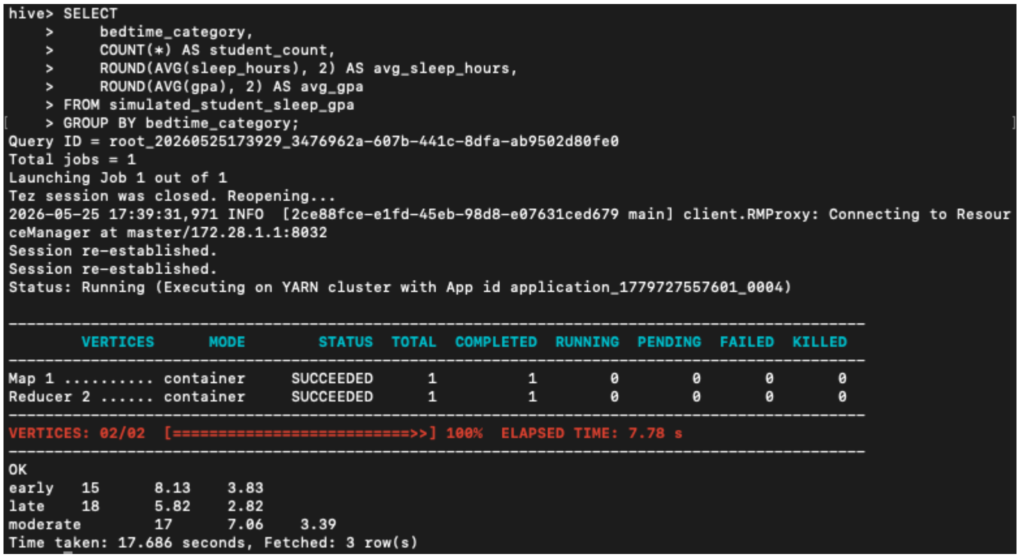
```
bash-5.0# hdfs dfs -ls /tmp
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/sl
f4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/tez/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBi
nder.class]
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLogg
erBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2026-05-25 17:06:55,696 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform...
using builtin-java classes where applicable
Found 2 items
drwx-wx-wx   - root           supergroup          0 2026-05-25 16:47 /tmp/hive
-rw-r--r--   3 charlottealley supergroup        918 2026-05-25 16:51 /tmp/simulated_student_sleep_gpa.csv
```

Hive Managed Table

The schema was designed to mimic the structure of the simulated student sleep data CSV file. INT was used for the student_id file because the column uses whole numbers. The sleep_hours and gpa columns were set to DOUBLE because they contained decimal values, similar to using the float data type in Python. The bedtime_category column uses STRING because it contains categorical text values. The table uses comma-separated values formatting because the original source is a CSV file, and the header row is skipped to avoid any errors with the inputted values.

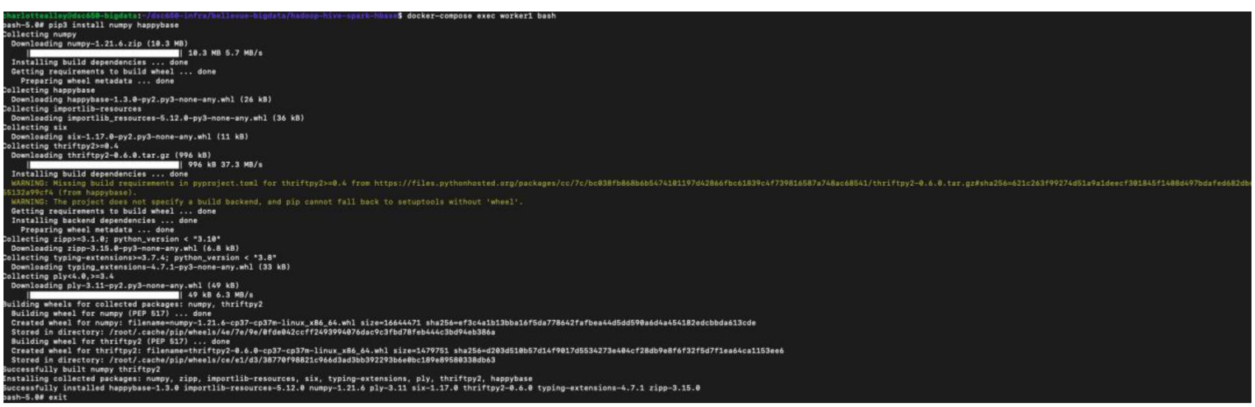
```
CREATE TABLE simulated_student_sleep_gpa (
  >    student_id INT,
  >    sleep_hours DOUBLE,
  >    gpa DOUBLE,
  >    bedtime_category STRING
  > )
  > ROW FORMAT DELIMITED
  > FIELDS TERMINATED BY ','
  > STORED AS TEXTFILE
  > TBLPROPERTIES ("skip.header.line.count"="1");
```

```
hive> LOAD DATA INPATH '/tmp/simulated_student_sleep_gpa.csv'
    > INTO TABLE simulated_student_sleep_gpa;
Loading data to table default.simulated_student_sleep_gpa
OK
Time taken: 1.183 seconds
hive> SELECT * FROM simulated_student_sleep_gpa LIMIT 10;
OK
1       6.2     3.1     late
2       7.5     3.6     moderate
3       5.8     2.9     late
4       8.1     3.8     early
5       6.9     3.3     moderate
6       7.2     3.5     moderate
7       5.4     2.7     late
8       8.4     3.9     early
9       6.7     3.2     moderate
10      7.9     3.7     early
Time taken: 3.474 seconds, Fetched: 10 row(s)
```



Environment Setup

The environment setup steps are necessary because the pipeline uses multiple big data tools, each serving its own purpose. By installing HappyBase, Python can connect to HBase through the Thrift interface and use its supporting packages (such as numpy). Together, these steps allow the data to move from file-based storage to queryable big data systems.



```
charlottealley@dsc650-bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ docker-compose exec worker2 bash
bash-5.0# pip3 install numpy happybase
Collecting numpy
  Downloading numpy-1.21.6.zip (10.3 MB)
     |████████████████████████████████| 10.3 MB 6.3 MB/s
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
    Preparing wheel metadata ... done
Collecting happybase
  Downloading happybase-1.3.0-py2.py3-none-any.whl (26 kB)
Collecting importlib-resources
  Downloading importlib_resources-5.12.0-py3-none-any.whl (36 kB)
Collecting six
  Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Collecting thriftpy2>=0.4
  Downloading thriftpy2-0.6.0.tar.gz (996 kB)
     |████████████████████████████████| 996 kB 26.3 MB/s
  Installing build dependencies ... done
  WARNING: Missing build requirements in pyproject.toml for thriftpy2>=0.4 from https://files.pythonhosted.org/packages/cc/7c/bc038fb868b6b5474101197d42866fbc61839c4f739816587a748ac68541/thriftpy2-0.6.0.tar.gz#sha256=621c263f99274d51a9a1deecf301845f1408d497bdafed682db61
55132a99cf4 (from happybase).
  WARNING: The project does not specify a build backend, and pip cannot fall back to setuptools without 'wheel'.
  Getting requirements to build wheel ... done
  Installing backend dependencies ... done
    Preparing wheel metadata ... done
Collecting zipp>=3.1.0; python_version < "3.10"
  Downloading zipp-3.15.0-py3-none-any.whl (6.8 kB)
Collecting typing-extensions>=3.7.4; python_version < "3.8"
  Downloading typing_extensions-4.7.1-py3-none-any.whl (33 kB)
Collecting ply<4.0,>=3.4
  Downloading ply-3.11-py2.py3-none-any.whl (49 kB)
     |████████████████████████████████| 49 kB 5.9 MB/s
Building wheels for collected packages: numpy, thriftpy2
  Building wheel for numpy (PEP 517) ... done
  Created wheel for numpy: filename=numpy-1.21.6-cp37-cp37m-linux_x86_64.whl size=16643805 sha256=8e7378b15e445102624d8c498a6f034163f76413e58660dae793fd36bf8f49c5
  Stored in directory: /root/.cache/pip/wheels/4e/7e/9e/0fde042ccff2493994076dac9c3fbd78feb444c3bd94eb386a
  Building wheel for thriftpy2 (PEP 517) ... done
  Created wheel for thriftpy2: filename=thriftpy2-0.6.0-cp37-cp37m-linux_x86_64.whl size=1479806 sha256=3512ae679b95c19ca9d5c936b7f80f42e994ee78a9b05f1847de283b3c109150
  Stored in directory: /root/.cache/pip/wheels/ce/e1/d3/38770f98821c966d3ad3bb392293b6e0bc189e89580338db63
Successfully built numpy thriftpy2
Installing collected packages: numpy, zipp, importlib-resources, six, typing-extensions, ply, thriftpy2, happybase
Successfully installed happybase-1.3.0 importlib-resources-5.12.0 numpy-1.21.6 ply-3.11 six-1.17.0 thriftpy2-0.6.0 typing-extensions-4.7.1 zipp-3.15.0
```

```
charlottealley@dsc650-bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ docker-compose exec master bash
bash-5.0# pip3 install numpy happybase
Collecting numpy
  Downloading numpy-1.21.6.zip (10.3 MB)
     |████████████████████████████████| 10.3 MB 6.0 MB/s
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
    Preparing wheel metadata ... done
Collecting happybase
  Downloading happybase-1.3.0-py2.py3-none-any.whl (26 kB)
Collecting importlib-resources
  Downloading importlib_resources-5.12.0-py3-none-any.whl (36 kB)
Collecting six
  Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Collecting thriftpy2>=0.4
  Downloading thriftpy2-0.6.0.tar.gz (996 kB)
     |████████████████████████████████| 996 kB 40.3 MB/s
  Installing build dependencies ... done
  WARNING: Missing build requirements in pyproject.toml for thriftpy2>=0.4 from https://files.pythonhosted.org/packages/cc/7c/bc038fb868b6b5474101197d42866fbc61839c4f739816587a748ac68541/thriftpy2-0.6.0.tar.gz#sha256=621c263f99274d51a9a1deecf301845f1408d497bdafed682db61
55132a99cf4 (from happybase).
  WARNING: The project does not specify a build backend, and pip cannot fall back to setuptools without 'wheel'.
  Getting requirements to build wheel ... done
  Installing backend dependencies ... done
    Preparing wheel metadata ... done
Collecting zipp>=3.1.0; python_version < "3.10"
  Downloading zipp-3.15.0-py3-none-any.whl (6.8 kB)
Collecting typing-extensions>=3.7.4; python_version < "3.8"
  Downloading typing_extensions-4.7.1-py3-none-any.whl (33 kB)
Collecting ply<4.0,>=3.4
  Downloading ply-3.11-py2.py3-none-any.whl (49 kB)
     |████████████████████████████████| 49 kB 5.8 MB/s
Building wheels for collected packages: numpy, thriftpy2
  Building wheel for numpy (PEP 517) ... done
  Created wheel for numpy: filename=numpy-1.21.6-cp37-cp37m-linux_x86_64.whl size=16644275 sha256=c6f2167af45810d24857606985e55d476f536952d4a2cbb7e54216e72e66049a
  Stored in directory: /root/.cache/pip/wheels/4e/7e/9e/0fde042ccff2493994076dac9c3fbd78feb444c3bd94eb386a
  Building wheel for thriftpy2 (PEP 517) ... done
  Created wheel for thriftpy2: filename=thriftpy2-0.6.0-cp37-cp37m-linux_x86_64.whl size=1479773 sha256=2ae08af34e0ed25c9b08b95d23c8f47f9920347b2040e244d1a5322934ca3045
  Stored in directory: /root/.cache/pip/wheels/ce/e1/d3/38770f98821c966d3ad3bb392293b6e0bc189e89580338db63
Successfully built numpy thriftpy2
Installing collected packages: numpy, zipp, importlib-resources, six, typing-extensions, ply, thriftpy2, happybase
Successfully installed happybase-1.3.0 importlib-resources-5.12.0 numpy-1.21.6 ply-3.11 six-1.17.0 thriftpy2-0.6.0 typing-extensions-4.7.1 zipp-3.15.0
bash-5.0#
```

```
bash-5.0# nohup hbase thrift start &
[3] 7763
bash-5.0# nohup: ignoring input and appending output to 'nohup.out'
jps
528 NameNode
2688 HistoryServer
1089 RunJar
1090 RunJar
1060 JobHistoryServer
2741 HMaster
662 SecondaryNameNode
7894 Jps
762 ResourceManager
7626 ThriftServer
2684 Master
[3]+  Exit 1                  nohup hbase thrift start
```

HBase Table Creation

The HBase row key was designed with the student_id in mind, because each student has a unique identifier in the synthetic dataset. This design makes it easy to look up each individual record. The table uses a single-column family to store each column, because all the fields describe the same student record and are likely to be accessed together. This keeps the HBase design simple while still supporting quick row-based lookups.

```
hbase(main):001:0> create 'my_table', 'cf'
Created table my_table
Took 1.8855 seconds
```

```
hbase(main):004:0> scan 'my_table'
ROW                           COLUMN+CELL
0 row(s)
Took 0.0663 seconds
```

PySpark ML Code

I chose to run a Linear Regression model because the target variable is both numeric and continuous (GPA). The model used sleep_hours and bedtime_category as input features to predict GPA. This model was appropriate because the goal was to estimate a continuous outcome rather than to classify students into categories. The RMSE shows the average size of the model's prediction error (0.066), while the R2 shows how much of the variation in GPA the model explains (.978). These results suggest a very strong fit and that sleep-related variables provide useful predictive information for student GPA (which is not surprising, given that simulated data was used).

```
from pyspark.sql import SparkSession
from pyspark.ml.feature import VectorAssembler, StringIndexer
from pyspark.ml.regression import LinearRegression
import happybase

# Step 1: Create a Spark session
```

```
spark = SparkSession.builder \
    .appName("Student Sleep GPA Prediction") \
    .enableHiveSupport() \
    .getOrCreate()

# Step 2: Load data from student sleep Hive table into Spark df
student_df = spark.sql("""
    SELECT
        student_id,
        sleep_hours,
        gpa,
        bedtime_category
    FROM simulated_student_sleep_gpa
""")

# Step 3: Drop null values
student_df = student_df.na.drop()

# Step 4: Convert bedtime_category into a numeric feature
indexer = StringIndexer(
    inputCol="bedtime_category",
    outputCol="bedtime_category_index",
    handleInvalid="skip"
)

indexed_df = indexer.fit(student_df).transform(student_df)

# Step 5: Assemble features
assembler = VectorAssembler(
    inputCols=["sleep_hours", "bedtime_category_index"],
    outputCol="features",
    handleInvalid="skip"
)

assembled_df = assembler.transform(indexed_df).select("features", "gpa")

# Step 6: Split data into training and testing sets
```

```
train_data, test_data = assembled_df.randomSplit([0.7, 0.3], seed=42)

# Step 7: Train Linear Regression model
lr = LinearRegression(labelCol="gpa", featuresCol="features")
lr_model = lr.fit(train_data)

# Step 8: Evaluate model
test_results = lr_model.evaluate(test_data)

print(f"RMSE: {test_results.rootMeanSquaredError}")
print(f"R^2: {test_results.r2}")

# Step 9: Write metrics to HBase
data = [
    ("metrics1", "cf:rmse", str(test_results.rootMeanSquaredError)),
    ("metrics1", "cf:r2", str(test_results.r2)),
]

def write_to_hbase_partition(partition):
    connection = happybase.Connection("master")
    connection.open()
    table = connection.table("my_table")

    for row in partition:
        row_key, column, value = row
        table.put(row_key, {column: value})

    connection.close()

rdd = spark.sparkContext.parallelize(data)
rdd.foreachPartition(write_to_hbase_partition)

# Step 10: Stop Spark
spark.stop()
```

https://github.com/charlottealley/bigdata

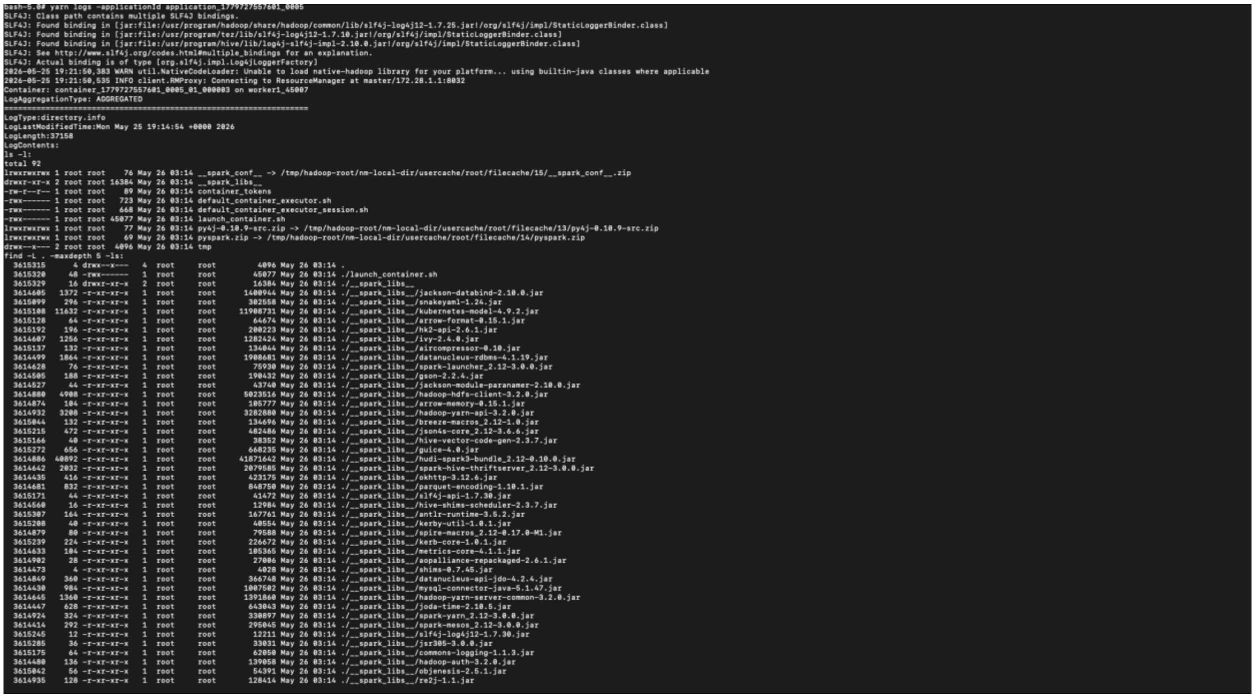
```
bash-5.0# spark-submit \
>   --master yarn \
>   --deploy-mode client \
>   --name StudentSleepGPA_to_HBase \
>   student_sleep_gpa_spark_hbase.py
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/spark/jars/slf4j-log4j12-1.7.30.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
0    [main] WARN  org.apache.hadoop.util.NativeCodeLoader  - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
1798 [Thread-5] INFO  org.apache.spark.SparkContext  - Running Spark version 3.0.0
1883 [Thread-5] INFO  org.apache.spark.resource.ResourceUtils  - ==============================================================
1887 [Thread-5] INFO  org.apache.spark.resource.ResourceUtils  - Resources for spark.driver:

1888 [Thread-5] INFO  org.apache.spark.resource.ResourceUtils  - ==============================================================
1889 [Thread-5] INFO  org.apache.spark.SparkContext  - Submitted application: Student Sleep GPA Prediction
2057 [Thread-5] INFO  org.apache.spark.SecurityManager  - Changing view acls to: root
2057 [Thread-5] INFO  org.apache.spark.SecurityManager  - Changing modify acls to: root
2057 [Thread-5] INFO  org.apache.spark.SecurityManager  - Changing view acls groups to:
2057 [Thread-5] INFO  org.apache.spark.SecurityManager  - Changing modify acls groups to:
2058 [Thread-5] INFO  org.apache.spark.SecurityManager  - SecurityManager: authentication disabled; ui acls disabled; users  with view permissions: Set(root); groups with view permissions: Set(); users  with modify permissions: Set(root); groups with modify permissions:
 Set()
2405 [Thread-5] INFO  org.apache.spark.util.Utils  - Successfully started service 'sparkDriver' on port 7001.
2496 [Thread-5] INFO  org.apache.spark.SparkEnv  - Registering MapOutputTracker
2599 [Thread-5] INFO  org.apache.spark.SparkEnv  - Registering BlockManagerMaster
2637 [Thread-5] INFO  org.apache.spark.storage.BlockManagerMasterEndpoint  - Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
2638 [Thread-5] INFO  org.apache.spark.storage.BlockManagerMasterEndpoint  - BlockManagerMasterEndpoint up
2722 [Thread-5] INFO  org.apache.spark.SparkEnv  - Registering BlockManagerMasterHeartbeat
2750 [Thread-5] INFO  org.apache.spark.storage.DiskBlockManager  - Created local directory at /tmp/blockmgr-dd5b2316-55c5-4416-be8a-6a71c80c2b9b
2811 [Thread-5] INFO  org.apache.spark.storage.memory.MemoryStore  - MemoryStore started with capacity 366.3 MiB
2885 [Thread-5] INFO  org.apache.spark.SparkEnv  - Registering OutputCommitCoordinator
3071 [Thread-5] INFO  org.sparkproject.jetty.util.log  - Logging initialized @5810ms to org.sparkproject.jetty.util.log.Slf4jLog
3210 [Thread-5] INFO  org.sparkproject.jetty.server.Server  - jetty-9.4.z-SNAPSHOT; built: 2019-04-29T20:42:08.989Z; git: e1bc35120a6617ee3df052294e433f3a25ce7097; jvm 1.8.0_275-b01
3244 [Thread-5] INFO  org.sparkproject.jetty.server.Server  - Started @5984ms
3285 [Thread-5] INFO  org.sparkproject.jetty.server.AbstractConnector  - Started ServerConnector@3ff5d4ba{HTTP/1.1,[http/1.1]}{172.28.1.1:4040}
3285 [Thread-5] INFO  org.apache.spark.util.Utils  - Successfully started service 'SparkUI' on port 4040.
3332 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@6ed7ab6a{/jobs,null,AVAILABLE,@Spark}
3337 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@7c6ad1cb{/jobs/json,null,AVAILABLE,@Spark}
3338 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@79221c6e{/jobs/job,null,AVAILABLE,@Spark}
3342 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@66960afc{/jobs/job/json,null,AVAILABLE,@Spark}
3343 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@65a6d8e9{/stages,null,AVAILABLE,@Spark}
3344 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@a65220{/stages/json,null,AVAILABLE,@Spark}
3345 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@298910ed{/stages/stage,null,AVAILABLE,@Spark}
3347 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@2f86a3f{/stages/stage/json,null,AVAILABLE,@Spark}
3349 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@1bbba702{/stages/pool,null,AVAILABLE,@Spark}
3350 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@11946de{/stages/pool/json,null,AVAILABLE,@Spark}
3351 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@70991c86{/storage,null,AVAILABLE,@Spark}
3353 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@706c5b7c{/storage/json,null,AVAILABLE,@Spark}
3354 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@1fd555af{/storage/rdd,null,AVAILABLE,@Spark}
3357 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@702e185b{/storage/rdd/json,null,AVAILABLE,@Spark}
3359 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@6253f6e0{/environment,null,AVAILABLE,@Spark}
3361 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@58c8e008{/environment/json,null,AVAILABLE,@Spark}
3363 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@7a195f0{/executors,null,AVAILABLE,@Spark}
3364 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@661095fc{/executors/json,null,AVAILABLE,@Spark}
3365 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@5356e470{/executors/threadDump,null,AVAILABLE,@Spark}
3367 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@7719e045{/executors/threadDump/json,null,AVAILABLE,@Spark}
3386 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@54389234{/static,null,AVAILABLE,@Spark}
3387 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@6a0a2dac{/,null,AVAILABLE,@Spark}
3389 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@59d187ac{/api,null,AVAILABLE,@Spark}
3390 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@6c747a1c{/jobs/job/kill,null,AVAILABLE,@Spark}
3391 [Thread-5] INFO  org.sparkproject.jetty.server.handler.ContextHandler  - Started o.s.j.s.ServletContextHandler@76c97444{/stages/stage/kill,null,AVAILABLE,@Spark}
3394 [Thread-5] INFO  org.apache.spark.ui.SparkUI  - Bound SparkUI to 172.28.1.1, and started at http://localhost:4040
4081 [Thread-5] INFO  org.apache.hadoop.yarn.client.RMProxy  - Connecting to ResourceManager at master/172.28.1.1:8032
4600 [Thread-5] INFO  org.apache.spark.deploy.yarn.Client  - Requesting a new application from cluster with 2 NodeManagers
4823 [Thread-5] INFO  org.apache.hadoop.conf.Configuration.deprecation  - No unit for dfs.client.datanode-restart.timeout(30) assuming SECONDS
5281 [Thread-5] INFO  org.apache.hadoop.conf.Configuration  - resource-types.xml not found
5282 [Thread-5] INFO  org.apache.hadoop.yarn.util.resource.ResourceUtils  - Unable to find 'resource-types.xml'.
5335 [Thread-5] INFO  org.apache.spark.deploy.yarn.Client  - Verifying our application has not requested more than the maximum memory capability of the cluster (1536 MB per container)
5335 [Thread-5] INFO  org.apache.spark.deploy.yarn.Client  - Will allocate AM container, with 896 MB memory including 384 MB overhead
5336 [Thread-5] INFO  org.apache.spark.deploy.yarn.Client  - Setting up container launch context for our AM
5338 [Thread-5] INFO  org.apache.spark.deploy.yarn.Client  - Setting up the launch environment for our AM container
5355 [Thread-5] INFO  org.apache.spark.deploy.yarn.Client  - Preparing resources for our AM container
```

```
bash-5.0# grep -i "RMSE\|R^2" spark_output.txt
RMSE: 0.06562225809386474
R^2: 0.9778711163548931
```

Spark Submit & Output

```
bash-5.0# spark-submit \
>   --master yarn \
>   --deploy-mode client \
>   --name StudentSleepGPA_to_HBase \
>   student_sleep_gpa_spark_hbase.py
```

```
        client token: N/A
        diagnostics: N/A
        ApplicationMaster host: 172.28.1.3
        ApplicationMaster RPC port: -1
        queue: root.root
        start time: 1779736432623
        final status: UNDEFINED
        tracking URL: http://master:8088/proxy/application_1779727557601_0005/
        user: root
```



HBase Scan Verification

The metrics written to HBase were RMSE and R2 from the PySpark Linear Regression model. As previously stated, RMSE measures the average prediction error for GPA, and R2 measures how well the model explains GPA variation. These results were successfully stored in HBase, validating that the pipeline worked end-to-end.

```
bash-5.0# hbase shell
scan 'my_table'
2026-05-25 19:39:32,924 WARN  [main] util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
HBase Shell
Use "help" to get list of supported commands.
Use "exit" to quit this interactive shell.
For Reference, please visit: http://hbase.apache.org/2.0/book.html#shell
Version 2.3.6, r7414579f2620fca6b75146c29ab2726fc4643ac9, Wed Jul 28 22:24:42 UTC 2021
Took 0.0022 seconds
hbase(main):001:0> scan 'my_table'
ROW                          COLUMN+CELL
 metrics1                    column=cf:r2, timestamp=2026-05-25T19:31:06.805, value=0.9778711163548931
 metrics1                    column=cf:rmse, timestamp=2026-05-25T19:31:06.823, value=0.06562225809386474
1 row(s)
Took 0.8711 seconds
hbase(main):002:0>
```

Conclusion

The data was ingested via NiFi, loaded into HDFS, loaded into a Hive table, analyzed in Spark using SQL and MLlib, and the final evaluation metrics were written to HBase for easy querying. This type of pipeline, when applied to real-world data, can yield insightful, data-driven results that guide business decisions in ways not possible before the introduction of big data tools.